

### 8.3. Adaptive Control of Linear Systems With Full State Feedback.

- Consider the adaptive control of  $n^{\text{th}}$ -order linear systems

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_0 y = u \quad (1)$$

- state components  $y, \dot{y}, \dots, y^{(n-1)}$  are measurable
- assume coefficient vector  $a = \begin{pmatrix} a_n \\ \vdots \\ a_1 \\ a_0 \end{pmatrix}$  is unknown, the sign of  $a_n$  is known.

Example: An example of such systems is the dynamics of a mass-spring-damper system

$$m\ddot{y} + c\dot{y} + ky = u.$$

Objective: make  $y$  closely track the response of a stable reference model

$$a_n y_m^{(n)} + a_{n-1} y_m^{(n-1)} + \dots + a_0 y_m = r(t) \quad (2)$$

with  $r(t)$  being a bounded reference signal.

Choice of control law:

- Define a signal  $z(t) = y_m^{(n)} - \beta_{n-1} e^{(n-1)} - \dots - \beta_0 e$  (3)

with  $\beta_1, \dots, \beta_n$  being positive constants chosen s.t.  $p^n + \beta_{n-1} p^{n-1} + \dots + \beta_0$  is a stable (Hurwitz) polynomial. i.e.  $1 > \beta_{n-1} > \dots > \beta_0 > 0$ .

- Adding  $(-a_n z(t))$  to both sides of (1) and rearranging, the plant dynamics as

$$a_n[y^{(n)} - z] = u - a_n z - a_{n-1} y^{(n-1)} - \dots - a_0 y \quad (4)$$

• Choose the control law to be

$$u = \hat{a}_n z + \hat{a}_{n-1} y^{(n-1)} + \dots + \hat{a}_0 y = v^T(t) \hat{a}(t) \quad (5) \quad \text{where } v(t) = \begin{pmatrix} z(t) \\ y^{(n-1)} \\ \vdots \\ \dot{y} \\ y \end{pmatrix}$$

$$\hat{a}(t) = \begin{pmatrix} \hat{a}_n \\ \vdots \\ \hat{a}_0 \end{pmatrix} \rightarrow \text{estimated parameter vector}$$

$\Rightarrow$  The tracking error  $e = y - y_m$  satisfies the closed-loop dynamics

$$a_n[e^{(n)} + \beta_{n-1} e^{(n-1)} + \dots + \beta_0 e] = v^T(t) \tilde{a}(t), \quad \text{where } \tilde{a} = \hat{a} - a. \quad (6)$$

[ Substituting (3), (5) into (4), then

$$\begin{aligned} & a_n[y^{(n)} - y_m^{(n)} + \beta_{n-1} e^{(n-1)} + \dots + \beta_0 e] \\ &= \hat{a}_n z + \hat{a}_{n-1} y^{(n-1)} + \dots + \hat{a}_0 y - a_n z - a_{n-1} y^{(n-1)} - \dots - a_0 y, \end{aligned}$$

i.e.

$$\begin{aligned} a_n[e^{(n)} + \beta_{n-1} e^{(n-1)} + \dots + \beta_0 e] &= (\hat{a}_n - a_n) z + (\hat{a}_{n-1} - a_{n-1}) y^{(n-1)} + \dots + (\hat{a}_0 - a_0) y \\ &= v^T(t) \tilde{a}(t). \end{aligned}$$

Choice of adaptation law

• Rewrite the closed-loop error dynamics (6) in state space form

$$\dot{x} = Ax + b \left[ \left( \frac{1}{a_n} \right) v^T \tilde{a} \right]$$

$$e = cx.$$

where

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -\beta_0 & -\beta_1 & -\beta_2 & \dots & -\beta_{n-1} \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}, \quad C = (1, 0, \dots, 0).$$

- Consider the Lyapunov function candidate

$$V(X, \hat{a}) = X^T P X + \hat{a}^T T^{-1} \hat{a}$$

where both  $T$  and  $P$  are symmetric positive definite constant matrices, and  $P$  satisfies

$$PA + A^T P = -Q, \quad Q = Q^T > 0 \quad \text{for a chosen } Q$$

- The derivative  $\dot{V}$  can be computed as

$$\dot{V} = -X^T Q X + 2\hat{a}^T v b^T P X + 2\hat{a}^T T^{-1} \dot{\hat{a}}$$

$\Rightarrow$  Therefore, the adaptation law  $\dot{\hat{a}} = -T v b^T P X$  leads to

$$\dot{V} = -X^T Q X.$$

$\Rightarrow$  Therefore, with the adaptive controller defined by control law (5) and adaptation law (7),  $e$  and its  $(n-1)$  derivatives converge to 0.

#### § 8.4. Adaptive Control of Linear Systems with Output Feedback.

- only output measurement, rather than full state feedback.
- A linear time-invariant system can be represented by the transfer function

$$W(p) = K_p \frac{Z_p(p)}{R_p(p)}, \quad \text{where } K_p \text{ is called the high-frequency gain.}$$

$$R_p(p) = a_0 + a_1 p + \dots + a_{n-1} p^{n-1} + p^n$$

$$Z_p(p) = b_0 + b_1 p + \dots + b_{m-1} p^{m-1} + p^m$$

- In our adaptive control problem, the coefficients  $a_i, b_j$ ,  $i = 0, 1, \dots, n-1$ ,  $j = 0, 1, \dots, m-1$

and the high frequency gain  $k_p$  are all assumed to be unknown.

- The desired performance is assumed to be described by a reference model with transfer function

$$W_m(p) = K_m \frac{Z_m}{R_m}$$

where  $Z_m, R_m$  are monic Hurwitz polynomials of degrees  $n_m$  and  $m_m$ .  $k_m$  is positive.

- In our treatment, we will assume  $n_m - m_m \geq n - m$ .

Objective The objective of the design is to determine a control law, and an associated adaptation law, so that the plant output  $y$  asymptotically approaches  $y_m$ .

#### 8.4.1. Linear Systems with Relative Degree One.

When the relative degree is 1, i.e.  $m = n - 1$ . (The degree of denominator is one order higher than that of numerator)

Example: • Consider the plant described by

$$y = \frac{k_p(p + b_p)}{p^2 + a_{p1}p + a_{p2}} u$$

and the reference model

$$y_m = \frac{k_m(p + b_m)}{p^2 + a_{m1}p + a_{m2}} r$$

- Let the controller with the control law being

$$u = \alpha_1 z + \frac{\beta_1 p + \beta_2}{p + b_m} y + k r$$

where  $z = \frac{n}{p + b_m}$ ,  $\alpha_1, \beta_1, \beta_2, k$  are controller parameters.

• If we take these parameters to be

$$\alpha_1 = b_p - b_m,$$

$$\beta_1 = \frac{\alpha_{m1} - \alpha_{p1}}{k_p},$$

$$\beta_2 = \frac{\alpha_{m2} - \alpha_{p2}}{k_p}$$

$$k = \frac{k_m}{k_p}$$

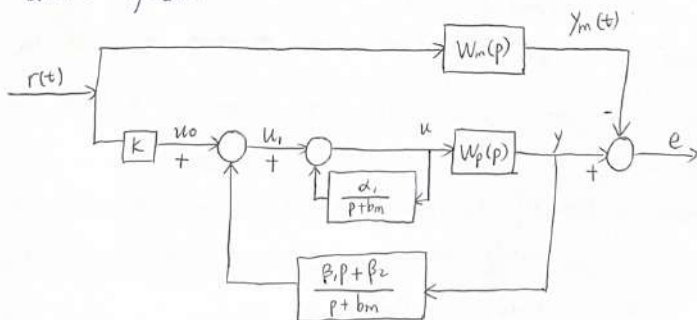
⇒ one can show that the transfer function from the reference input  $r$  to the plant output  $y$  is

$$W_{ry} = \frac{k_m(p + b_m)}{p^2 + a_m p + \alpha_{m2}} = W_m(p) \quad \text{I have no idea how to calculate this}$$

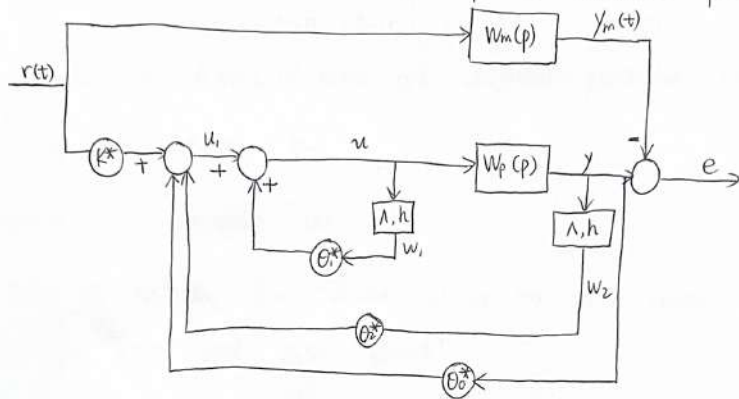
⇒ perfect tracking is achieved with this control law, i.e.  $y(t) = y_m(t)$ .

(What does this mean?)

Control system:



- The controller structure for second-order plants can be extended to any plant with relative degree one. The resulting structure is the following, where  $k^*$ ,  $\theta_1^*$ ,  $\theta_2^*$  and  $\theta_0^*$  represents controller parameters.



- The block for generating the filter signal  $w_1$  represents an  $(n-1)^{th}$  order dynamics,

$$\dot{w}_1 = \Lambda w_1 + h u \quad \text{where } w_1 \text{ is an } (n-1) \times 1 \text{ state vector,}$$

$\Lambda$  is an  $(n-1) \times (n-1)$  matrix, and

$h$  is a constant.

$$\det(pI - \Lambda) = Z_{n-1}(p)$$

- The block for generating the  $(n-1) \times 1$  vector  $w_2$  has the same dynamics but with  $y$  as input, i.e.

$$\dot{w}_2 = \Lambda w_2 + h y$$

- The scalar gain  $k^*$  is defined to be  $k^* = \frac{k_m}{k_p}$  and is intended to modulate the high-frequency gain of the control system.
- The vector  $\theta_1^*$  contains  $(n-1)$  parameters which intend to cancel the zeros of the plant.
- The vector  $\theta_2^*$  contains  $(n-1)$  parameters which, together with the scalar

gain  $\theta_0^*$  can move the poles of the closed-loop control system to the locations of the reference model poles.

$\Rightarrow$  To this end, the control law is chosen to be

$$u = k(t)r + \theta_1(t)w_1 + \theta_2(t)w_2 + \theta_0(t)y, \quad (8)$$

where  $k(t)$ ,  $\theta_1(t)$ ,  $\theta_2(t)$ ,  $\theta_0(t)$  are controller parameters to be provided by the adaptation law.

Choice of adaptation law

- Let  $\theta$  be the  $2 \times 1$  vector,  $w$  be the  $2 \times 1$  vector, i.e.

$$\theta(t) = [k(t) \quad \theta_1(t) \quad \theta_2(t) \quad \theta_0(t)]^T$$

$$w(t) = [r(t) \quad w_1(t) \quad w_2(t) \quad y(t)]^T$$

Then the control law (8) can be compactly written as

$$u = \theta^T(t)w(t) \quad (9)$$

- Denote the ideal value of  $\theta$  by  $\theta^*$

the error between  $\theta(t)$  and  $\theta^*$  by  $\phi(t) = \theta(t) - \theta^*$ ,

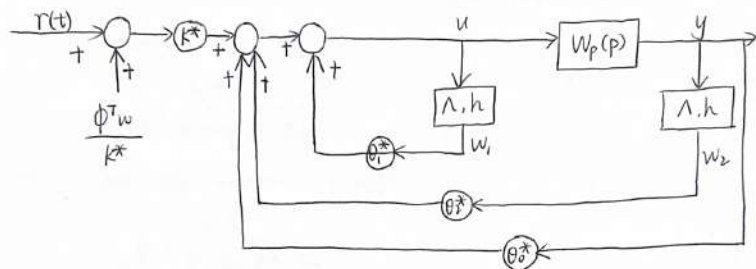
the estimated parameters  $\theta(t)$  can be represented as  $\theta(t) = \theta^* + \phi(t)$ .

Then the control law (8) can be written as

- $$u = (\theta^*)^T w + \phi^T(t) w.$$

- In order to choose an adaptation law so that the tracking error  $e$  converges to zero, we have to first find out how the tracking error is related to the parameter error.

- With the control law given in (9), the control system with variable gains can be represented as follows, with  $\frac{\phi^T(t)w}{k^*}$  regarded as an external signal.



the output here must be

$$y(t) = W_m(p)r + W_m(p) \cdot \frac{\phi^T(t)w}{k^*}$$

Since  $y_m(t) = W_m(p)r$ , the tracking error is seen to be related to the parameter error by

$$e(t) = W_m(p) \cdot \frac{\phi^T(t)w(t)}{k^*}$$

- According to lemma 8.1. the following adaptation law is chosen

$$\dot{\theta} = -\text{sgn}(k_p) \gamma e(t) w(t)$$

#### § 8.4.2. Linear Systems with Higher <sup>relativ</sup> Relative Degree.

- The design of adaptive controller for plants with relative degree larger than 1 is both similar to, and different from, that for plants with relative degree 1.



- Specifically, the choice of control law is quite similar but the choice of adaptation law is very different.

Choice of control law

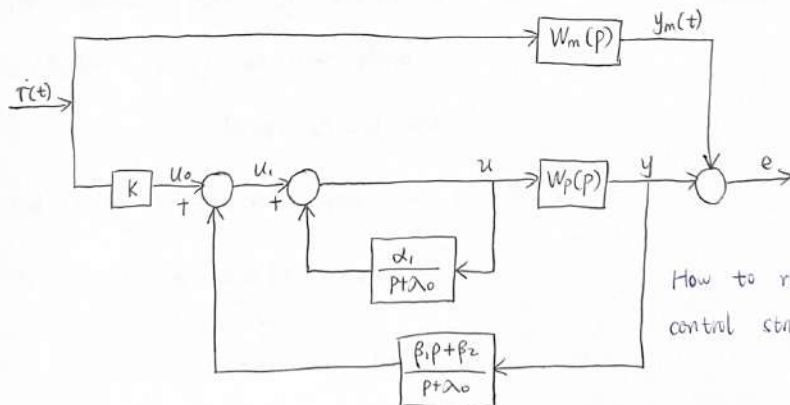
Example: Consider the second order plant described by the transfer function

$$y = \frac{k_p u}{p^2 + a_{p1}p + a_{p2}}$$

and the reference model

$$y_m = \frac{k_m r}{p^2 + a_{m1}p + a_{m2}}$$

- Let us consider the control structure shown as follows.



How to read this control structure?

- The closed-loop transfer function from the reference signal  $r$  to the plant output  $y$  is

$$W_{ry} = k \frac{\frac{p+\lambda_0}{p+\lambda_0+\alpha_1} \frac{k_p}{p^2+a_{p1}p+a_{p2}}}{1 + \frac{p+\lambda_0}{p+\lambda_0+\alpha_1} \frac{\beta_1 p + \beta_2}{p+\lambda_0} \frac{k_p}{p^2+a_{p1}p+a_{p2}}} = \frac{k k_p (p+\lambda_0)}{(p+\lambda_0+\alpha_1)(p^2+a_{p1}p+a_{p2}) + \frac{k_p(\beta_1 p + \beta_2)}{p+\lambda_0}}$$

$\Rightarrow$  Therefore, if the controller parameters  $\alpha_1, \beta_1, \beta_2$  and  $k$  are chosen such that

$$(p + \lambda_0 + \alpha_1)(p^2 + \alpha_{p1}p + \alpha_{p2}) + k_p(\beta_1 p + \beta_2) = (p + \lambda_0)(p^2 + \alpha_{m1}p + \alpha_{m2})$$

$$k = \frac{k_m}{k_p},$$

$$\text{then } W_{ry} = \frac{\frac{k_m}{k_p} \cdot k_p \cdot (p + \lambda_0)}{(p + \lambda_0)(p^2 + \alpha_{m1}p + \alpha_{m2})} = \frac{k_m}{p^2 + \alpha_{m1}p + \alpha_{m2}}$$

i.e. the closed-loop transfer function  $W_{ry}$  becomes identically the same as that of the reference model. Clearly, such choice of parameters exists and is unique.

- For general plants of relative degree larger than 1, instead of  $\det[pI - A] = Z_m(p)$ , we now choose

$$\det[pI - A] = Z_m(p)\lambda_1(p)$$

and  $\lambda_1(p)$  is a Hurwitz polynomial of degree  $(n-1-m)$ .

- The closed-loop transfer function is

$$W_{ry} = \frac{k k_p z_p \lambda_1(p) Z_m(p)}{R_p(p)[\lambda(p) + c(p)] + k_p z_p D(p)}$$

Now, the question is whether in this general case there exist choice of values for  $k, \theta_0, \theta_1$  and  $\theta_2$  such that the above transfer function becomes exactly the same as  $W_m(p)$ , or equivalently,

$$R_p(\lambda(p) + c(p)) + k_p z_p D(p) = \lambda_1 z_p R_m(p). \quad (10)$$

- The answer to this question can be obtained from the following

Lemma: Let  $A(p)$  and  $B(p)$  be polynomials of degree  $n_1$  and  $n_2$ .

If  $A(p)$  and  $B(p)$  are relatively prime, then there exist polynomials  $M(p)$  and  $N(p)$  such that

$$\underline{A(p)} M(p) + \underline{B(p)} N(p) = \underline{A^*(p)},$$

where  $A^*(p)$  is an arbitrary polynomial.

- This lemma can be used straightforwardly to answer our question regarding (10).  $\underline{P_p} (\lambda(p) + C(p)) + \underline{k_p Z_p} D(p) = \underline{\lambda_1 Z_p R_m(p)}$

By regarding  $P_p$  as  $A(p)$ ,  $k_p Z_p$  as  $B(p)$  and  $\lambda_1 Z_p R_m$  as  $A^*(p)$ , we conclude that there exist polynomials  $\lambda(p) + C(p)$  and  $D(p)$  s.t. (10) is satisfied.

### Choice of Adaptation law

- We again use a control law of the form

$$u = \theta^T(t) w(t)$$

Using a similar reasoning as before, we can again obtain the output  $y$  in the form of  $y(t) = W_m(p)r + W_m(p)[\phi^T(t)w/k^*]$  and the tracking error is in the form of

$$e(t) = W_m(p)[\phi^T w/k^*] \quad (11)$$

- However, the choice of adaptation law given by  $\dot{\theta} = -\text{sgn}(k_p) r e(t) w(t)$  cannot be used. (Unlike to the case of plant with relative degree 1) because now the reference model transfer function  $W_m(p)$  is no longer SPR.

- A famous technique called error augmentation <sup>误差增强</sup> can be used to avoid the difficulty in finding an adaptation law for  $e(t) = W_m(p) [\phi^T(t) w] / k^*$ .
- Specifically, let us define an auxiliary error  $\eta(t)$  by   
 <sup>辅助误差</sup>

$$\eta(t) = \theta^T(t) W_m(p) [w] - W_m(p) [\theta^T(t) w(t)] \quad (12)$$

and define an augmented error  $\varepsilon(t)$  by combining the tracking error  $e(t)$  with the auxiliary error  $\eta(t)$  as

$$\varepsilon(t) = e(t) + \alpha(t) \eta(t) \quad (13)$$

~~For~~ Substituting (10), (12) into (13), we obtain

$$\varepsilon(t) = \frac{1}{k^*} \phi^T(t) \bar{w} + \phi_\alpha \eta(t),$$

$$\text{where } \phi_\alpha = \alpha(t) - \frac{1}{k^*}, \quad \bar{w}(t) = W_m(p) [w].$$

- A number of standard techniques to be discussed in §8.7 can be used to demonstrate that the controller parameters  $\theta(t)$  and the parameter  $\alpha(t)$  for forming the augmented error are updated by

$$\dot{\theta} = - \frac{\text{sgn}(k_p) \gamma \varepsilon \bar{w}}{1 + \bar{w}^T \bar{w}}, \quad \dot{\alpha} = - \frac{\gamma \varepsilon \eta}{1 + \bar{w}^T \bar{w}} \quad (14)$$

- With the control law  $u = \theta^T(t) w(t)$  and adaptation law (14), global convergence of the tracking error can be shown.

## § 8.5. Adaptive Control of Nonlinear Systems.

**Problem Statement** We consider  $n^{\text{th}}$ -order nonlinear systems

$$y^{(n)} + \sum_{i=1}^{n-1} d_i f_i(x, t) = bu, \quad (15)$$

where  $x = [y, \dot{y}, \dots, y^{(n-1)}]^T$  is the state vector

$f_i$  are known nonlinear functions of the state and time.

and the parameters  $\alpha_i$  and  $b$  are unknown constants. We assume that the state is measured, and that the sign of  $b$  is known. One example of such dynamics is

$$m\ddot{x} + c f_1(\dot{x}) + k f_2(x) = u,$$

which represents a mass-spring-damper system with nonlinear friction and nonlinear damping.

- The objective of the adaptive control design to make the output asymptotically tracks a desired output  $y_d(t)$  despite the parameter uncertainty
- Let us rewrite (15) as

$$h y^{(n)} + \sum_{i=1}^n \alpha_i f_i(x, t) = u, \quad (16)$$

by dividing both sides by the unknown constant  $b$ , where  $h = \frac{1}{b}$  and  $\alpha_i = \frac{\alpha_i}{b}$

### choice of control law

- define a combined error

$$s = e^{(n-1)} + \lambda_{n-2} e^{(n-2)} + \dots + \lambda_0 e \quad \text{~~define~~}$$

Note that  $s$  can be rewritten as

$$s = y^{(n-1)} - y_r^{(n-1)},$$

↓

$$y_d^{(n-1)} - \lambda_{n-2} e^{(n-2)} - \dots - \lambda_0 e.$$

- Consider the control law

$$u = h y_r^{(n)} - k s + \sum_{i=1}^n \alpha_i f_i(x, t), \quad (17)$$

where  $k$  is a constant of the same sign as  $h$ ,

$$y_r^{(n)} = y_d^{(n)} - \lambda_{n-2} e^{(n-2)} - \dots - \lambda_0 \dot{e}$$

- Substituting (17) into (16), we see

$$h y_r^{(n)} + \sum_{i=1}^n a_i f_i(x, t) = h y_r^{(n)} - k s + \sum_{i=1}^n a_i f_i(x, t)$$

$$h \dot{s} + k s = 0$$

and therefore gives exponential convergence of  $s$ , which, in turn, guarantees the convergence of  $e$ .

Choice of adaptation law

- For our adaptive control, the control law (17) is replaced by

$$u = \hat{h} y_r^{(n)} - k s + \sum_{i=1}^n \hat{a}_i f_i(x, t),$$

where  $h$  and the  $a_i$  have been replaced by their estimated values. The tracking error from this control law can be easily shown to be

$$h \dot{s} + k s = \tilde{h} y_r^{(n)} + \sum_{i=1}^n \tilde{a}_i f_i(x, t).$$

- This can be rewritten as

$$s = \frac{1/h}{p + (k/h)} \left[ \tilde{h} y_r^{(n)} + \sum_{i=1}^n \tilde{a}_i f_i(x, t) \right].$$

Lemma 8.1. suggests us to choose the following adaptation law

$$\dot{\hat{h}} = -\gamma \operatorname{sgn}(h) s y_r^{(n)}$$

$$\dot{\hat{a}}_i = -\gamma \operatorname{sgn}(h) s f_i$$